

What does it take for a movie to be successful?

A case study using the IMDb metrics and data

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Abstract—The Internet Movie Database (IMDb) combines data from the most known movies on Earth. Using the data provided by this service, we developed a dashboard that shows to be effective and helpful in the task of finding what it takes for a movie to be successful.

Index Terms—IMDb, Data Visualization, Statistics, Shiny, Cinema, Movies, Crew, Decision Making

I. INTRODUCTION

Cinema, the seventh art, is an art form that paved its way in our lives during the twentieth century. Its growth, in a simplistic way, has been accompanied by the growth of Data Science and Statistics. This led to the creation and development of some tools and databases that combine all the fields mentioned above. One of these examples is the Internet Movie Database (IMDb) [1], which gathers several metrics regarding the most famous and acclaimed movies on Earth.

The aim of this project is to, using a dataset of the IMDb, determine which parameters lead a movie to success [6]. So, put in a even more simple way: "What is needed for a film to be successful?"

II. MATERIALS AND METHODS

A. The Initial Dataset

The initial dataset was provided to us, courtesy of the Professor, and it includes data from 5487 movies. All the movies are contemporary pieces, being launched between 2000 and 2020. In order for us to accomplish the mission, the dataset contains several parameters that describe all the movies: 'imdb title id', 'title', 'original title', 'year', 'date published', 'genre', 'duration', 'country', 'language 1', 'language 2', 'language 3', 'director', 'writer', 'actors', 'actors 1', 'actors f2', 'description', 'desc35', 'avg vote', 'votes', 'budget', 'USA gross income', 'worldwide gross income', 'reviews from users'.

In the preliminary analysis report, we went further on the definition and meaning of each parameter. In short, they describe the movie with biographic, financial, cast, public acclamation and synopsis parameters.

B. Data Pre-Processing

During the Exploratory Data Analysis (EDA), we encountered several issues in the data quality that required our attention. The first problem we encountered were 'numerical' variables misclassified as 'character' variables, namely in the

columns 'worldwide gross income' and 'budget'. There were also rows containing missing values for these attributes and we decided to drop these cases for lack of viable imputation strategies. Furthermore, we found that the remaining financial attribute, 'usa gross income', provides a narrowed and possibly even skewed distribution of the overall earnings of the movies so we decided to drop it. Lastly, the financial variables were converted into the same currency (USD) and combined to create a new variable that will define financial and commercial success, 'profit', by subtracting the budget from the overall earnings [2].

Missing values were also found in high numbers for the variables 'language 2' and 'language 3', which we have deemed irrelevant for this project and thus decided not to keep them in the dataset.

Another issue arose with outliers in the 'date published' attribute, which spanned back to 1905, indicating incorrect year, month and day values in seven rows, which we have decided to simply drop since it may impact our temporal analysis. Further addressing the temporal attributes, we recognized the redundancy of keeping the variable 'year' (referring to year of production) in our dataset since 'date published' also contains the year of publishing and in most cases the year of production is the same. Furthermore, in the cases where the years of production and publication do vary, the year of publication is more important and accurate for temporal analysis of trends and patterns. For this reason, we have decided to discard the 'year' variable.

Upon inspecting the movie titles, we realized there are no repeated movie titles. For this reason, we have decided to discard the 'title' variable along with the 'imdb title id' attribute and keep only the 'original title' as the unique identifier for our samples.

Since 'actors' already includes the star of the movie ('actors 1') and the co-star (second name in the 'actors f2' attribute), these variables become redundant. For this reason, we excluded both 'actors 1' and 'actors f2' and kept the variable with information regarding the whole cast ('actors') in play since it is already in order of importance (star & co-star & secondary cast).

Lastly, the attribute 'desc35' represents only the first 35 characters in the description. This could potentially be informative, but we have found it to be a brief and redundant

summary of 'description'. With that in mind, we have also decided to drop this variable.

C. Exploratory Data Analysis (EDA)

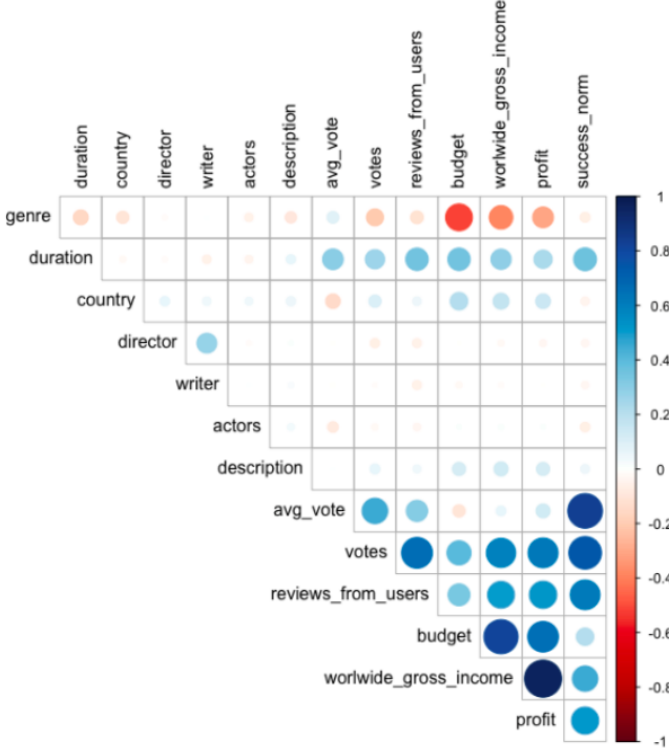


Fig. 1. Correlation matrix between the parameters in the dataset.

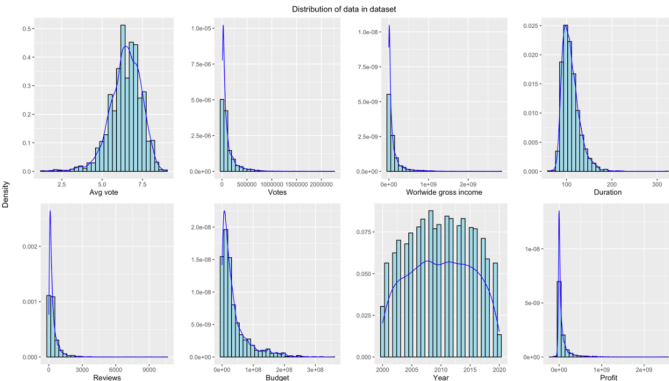


Fig. 2. Data distribution during the Exploratory Data Analysis.

III. THE FORMULA TO SUCCESS

$$\text{Success} = (\mu\text{Vote} + \text{Profit}) + \left(\frac{\text{N.}^\circ \text{ of votes} + \text{N.}^\circ \text{ of reviews from users}}{1 + (\text{Age}/100)} \right) \quad (1)$$

IV. RESULTS AND DISCUSSION

A. Dashboard

1) *User input:* To determine and answer the main question of what it takes for a movie to be successful, we built a dashboard using shiny [3]–[5] which, intuitively, guides the user to explore and discover which values in the parameters are more effective to the success outcome of an hypothetical movie. These statistics and visualizations are based on the data we gathered in our success subset, or in the initial dataset.

Fig. 3. The input section in the dashboard.

The user, after selecting the desired dataset/subset, may choose the genre of the movie under study. The logic behind this decision is because the genre parameter is one of the most diverse in the data, but not too sparse, thus being the perfect starting point to surf the data looking for the right features to a successful movie. Being a dynamic dashboard, the type of visualization and the data in it are dependent of the selections made by the user.

Besides the genre parameter, we give the user the option to go further and filter the search to actors and directors. In a decision making point of view, useful for a cinema production company, the casting is a step almost as important as choosing the genre, with the ultimate goal of driving a movie to success. Other parameters could be inserted to be filtered by the user, but we believe it would lead to a more confusing user experience.

2) *Dashboard output:* The dashboard output is composed by five tabs: 'Genre', 'Duration', 'Crew', 'Financial' and 'Trends'.

The '**Genre**' tab contains three main visualizations, aiming to give a more clear theme to the movie. This can be checked

in a word cloud with the most common words in the selected genre and crew. It also shows several combinations of genres, sorted by frequency and success. Finally, a heatmap gives a better understanding of when the movie should be released.

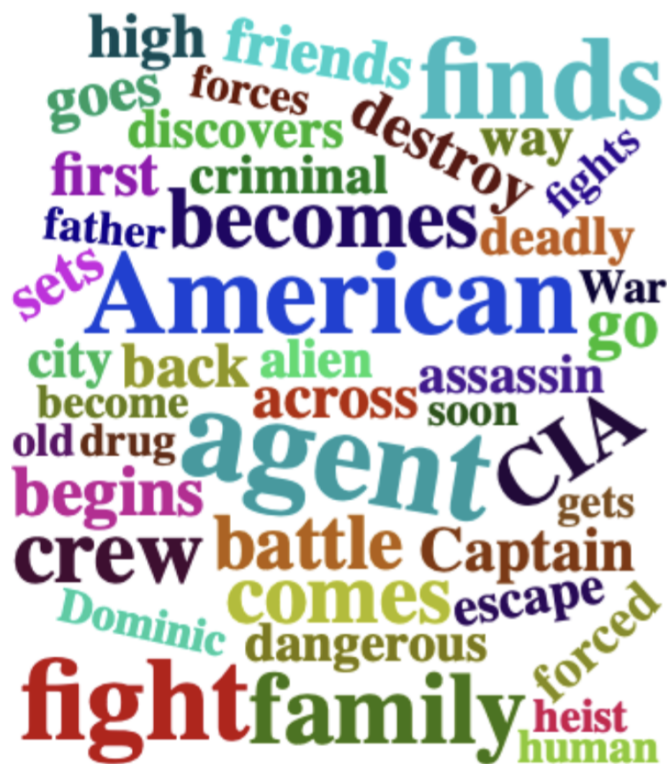


Fig. 4. The Word Cloud of genre 'Action'.

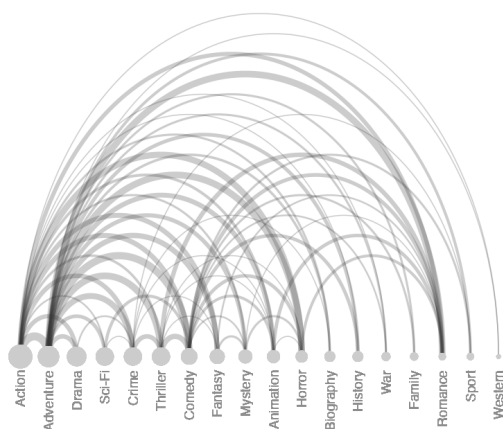


Fig. 5. Relations between movie genres.

The **Duration** tab takes a simpler approach, showing the recommended duration for a movie with the selected filters, in order to increase its chances of success.

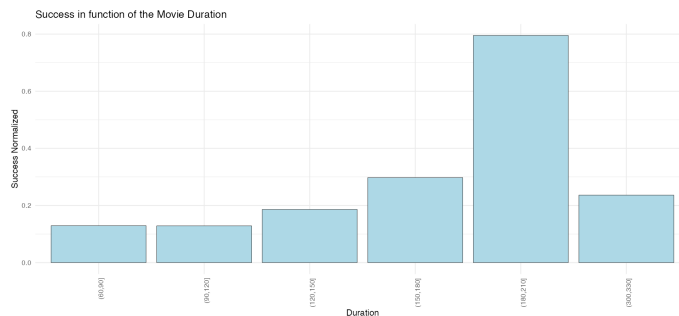


Fig. 6. Histogram relating the movie duration and the success normalized for the genre 'Action'.

The casting, as mentioned before, is one of the major steps that a production company must do while producing a movie. The **Crew** tab in our dashboard aims to help doing the right choices. After selecting the desired genre and/or crew, the production company will be guided by the dashboard with suggestions that optimise the likelihood of success. So, it will suggest the best actors to fill a role in the movie, the best directors (in case he/she/they weren't selected before)... And, finally, it will provide a network containing the most successful relations between directors and actors.

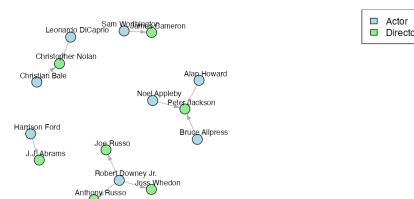


Fig. 7. Most successful relations between 'Actors' and 'Directors' in the genre 'Action'.

The **Financial** tab provides an overview to several financial parameters in the dataset, as 'Profit', 'Worldwide Income', 'Budget'; and critical parameters as 'Average Vote', 'Number of votes' and 'Reviews from users'. This insight provides an important message to the production company, considering they have to define how much money and resources will be spend in the movie.

The final tab in our dashboard is the **Trends** tab. How successful, in the next years, will a genre, director, or actor be? Is it expected for a genre to continue being profitable in the future? How much will the income grow? Should we invest more money, to keep up with our competitors? And what about critical acclamation, what are the expectations? Our final tab answers to all these questions. It takes into account the past events in our data, their trend through time, and it tries to predict the future of movies and cinema.

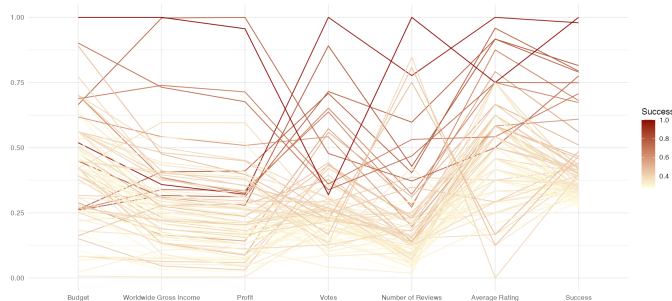


Fig. 8. Parallel plot with financial and critical parameters regarding the genre 'Action'.

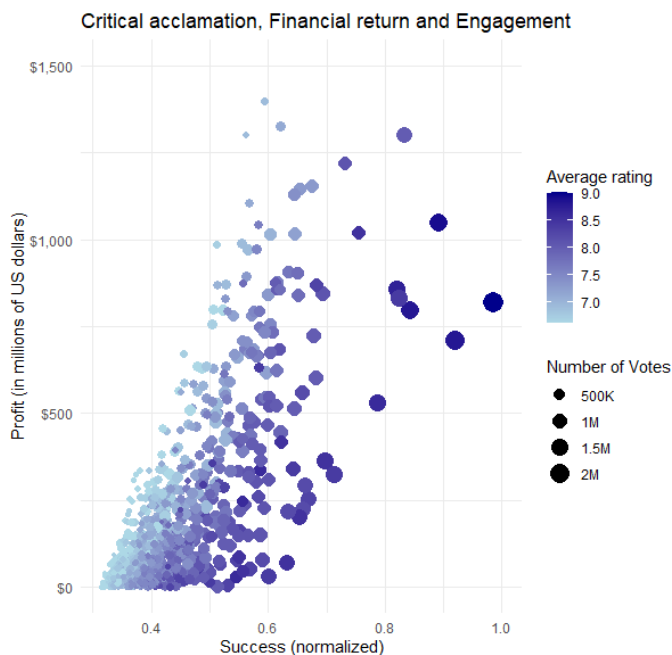


Fig. 9. Scatter plot relating 'Success', 'Avg Vote', 'Number of Votes' and 'Profit' for the genre 'Action'.

B. Dashboard to Success: A Case Study

The dashboard is a highly interactive and dynamic tool. So, to illustrate the way our dashboard works, we will describe a case study. Let's take, for example, the genre 'Action' as our choice.

The first information provided by the dashboard regards the combination of genres. In our data, the combination between the genre 'Action' and 'Adventure' has the highest average success value: 0,594. So, if we were in doubt regarding the second genre of our future production, 'Adventure' could be a safe bet.

Secondly, the theme of the movie can be better understood with the help of the word cloud provided by the dashboard: 'agent', 'American', 'fight', 'CIA', 'family', 'criminal' are very common words in our most successful movies. This, while expected, shows that Hollywoodian productions are the safe bet in this genre.

Regarding the best time of the year to release the movie, our tool suggests that the summer time should be the best time to do it, in order to increase the chances of success. It also provides an insight on the duration of the movie, as the most successful movies in the category had a duration between 180 and 210 minutes.

In the crew segment, we got suggestions regarding the best main actors for the movie. Leonardo DiCaprio is the golden selection in this case, but Noel Appleby and Alan Howard are very close. Regarding the choice of director, Joe Russo and Anthony Russo are great choices. The complementary visualization in our 'Crew' tab shows that Robert Downey Jr has a great relationship between both Anthony Russo and Joe Russo. So, even that Robert Downey Jr was not suggested as the top actor for this genre, if we choose one of the two main suggested directors, it is also a great idea to include Robert Downey Jr in the cast. Specially if we consider that, as shown in the first visualization, he also has a great performance and success in 'Action' movies.

Is it necessary to spend billions of USD to produce a great 'Action' movie? Well, our data, in the Financial tab, shows that it is usually the case... However, it also shows that the most successful movies in this genre are not, in all occasions, the ones who end up being highly profitable.

Finally, is it expected that the 'Action' genre will continue to grow its popularity and success in the next few years? The data shows that it may enter a plateau period until 2025. On the other hand, the income is expected to continue the tendency to grow.

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